

Spectral-folding model and capacity boundary for Laguerre–Gaussian mode sorting in Fabry–Pérot cavities

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Abstract: Fabry–Pérot (FP) cavities provide a state-preserving platform for sorting orbital angular momentum (OAM) states of light, with applications in optical communications and photonic quantum information processing. However, a general analytical model for high-dimensional Laguerre–Gaussian (LG) mode sorting in FP cavities has been lacking. Here, we present an analytical model based on spectral folding in FP cavities. By mapping folded resonance positions onto a single free spectral range (FSR) circle, the cavity design problem is reduced to maximizing the minimum circular spacing between target modes. A unified performance criterion then links this spacing and the cavity finesse to practical sorting metrics such as sorting efficiency and extinction ratio, establishing a capacity boundary for arbitrary finite target sets under a given performance threshold. The bound is tight for uniform-step target sets, for which the optimal cavity geometry is derived in closed form; for general nonuniform sets, the design reduces to an exact finite search over a bounded set of rational candidates. To experimentally validate the model, we examine a representative boundary case involving nine consecutive LG modes. Experimental measurements confirm the predicted folded resonance ordering, where the next mode returns to the fundamental’s spectral neighborhood. For an equal-weight nine-mode superposition state, the sorting achieves an average efficiency of 93.19% and a mean extinction ratio of 11.41 dB across all target modes, closely matching theoretical predictions.

1. Introduction

LG modes carrying OAM provide a natural high-dimensional degree of freedom for photonic quantum information processing [1, 2]. Their high dimensionality enhances the information capacity of photonic encoding, improves noise resilience in quantum communication, and facilitates high-dimensional entanglement as well as high-dimensional gate operations [3–6]. To harness these advantages practically, it is essential to spatially separate selected OAM modes for subsequent routing, interference, and quantum operations.

Over the past two decades, a variety of methods have been developed for sorting OAM modes. Leach et al. demonstrated an OAM beam splitter (OAM-BS) for odd–even sorting [7], and Slussarenko et al. subsequently extended this interferometric approach to a polarizing Sagnac configuration in spin–orbit photon processing [8]. Berkhout et al. proposed a log-polar coordinate transformation that maps OAM to position [9], and Mirhosseini et al. further improved its separation efficiency [10]. Subsequent advances along this transformation-based approach enhanced the sorting resolution [11] and enabled the modulo sorting of OAM modes [12]. More general approaches to spatial mode transformation have also emerged. Notably, multi-plane light conversion (MPLC) establishes a flexible platform for sorting modes across a wider range of spatial-mode bases [13] and has since been extended to the LG basis [14]. More recently,

FP-based approaches have been demonstrated for OAM-selective operations. Specifically, Wei et al. realized active sorting using cascaded tunable resonators [15], Vanani et al. extended FP-based mode selectivity to low-crosstalk mode-group demultiplexing with thin-film filters [16], and Yang et al. employed FP cavities as OAM-selective elements within an interferometric architecture for cyclic transformations [17].

These developments establish multiple technical routes for OAM discrimination. But their roles in quantum information processing are not identical. In practical scenarios, once high-dimensional quantum superposition or entangled states are generated, they necessitate both mode sorting and logic gate operations. Critically, the output states following sorting should preserve their original transverse spatial mode, amplitude and phase to facilitate further operations [6, 18]. In this context, state preservation constitutes a core requirement for OAM sorting in the field of OAM-based optical quantum information processing.

Among existing approaches, an early state-preserving scheme is the interferometric OAM-BS. This device utilizes OAM-dependent phase shifts induced by Dove prisms in a Mach–Zehnder (M-Z) interferometer to route different OAM components to separate output ports while maintaining their original state form [7, 8]. However, its limitation lies in its scalability to higher dimensions: accommodating an increasing number of target modes necessitates cascading additional M-Z interferometers, leading to a substantial rise in component count and imposing rigorous demands on phase stability. In contrast, FP cavities provide a different route to state-preserving OAM sorting. By exploiting Gouy-phase-dependent resonance [19, 20], an FP cavity achieves spectral-domain selection of transverse modes while leaving both the transmitted and reflected fields in their original state forms. Furthermore, FP-based OAM sorting is inherently dynamically tunable and modular: the target mode can be reconfigured by varying the cavity length, and the mode count can be scaled by appending modules at the cascade output without affecting preexisting ones [15]. These features make FP cavities attractive for coherent routing of OAM quantum states.

Existing FP-based studies have validated the feasibility of state-preserving OAM mode-selective operations in several scenarios [15–17]. However, these demonstrations have largely been confined to low-order or application-specific regimes and have not yielded a general design method for a chosen target-mode set with desired sorting performance.

In this work, we develop an analytical design model for high-dimensional LG mode sorting in FP cavities. The model addresses three practical design questions: how to choose the cavity geometry for a given target-mode set, how much finesse is required to achieve a desired sorting performance, and how many modes can be supported before spectral overlap becomes the limiting factor. Built on a spectral-folding description of cavity resonances, the model yields a capacity rule, closed-form optimal solutions for uniform-step target sets, and an exact search method for general finite sets. To validate the theory, we use the model to design and fabricate an FP cavity for a nine-mode target set and experimentally test its predictions.

2. Spectral-Folding Model and Design Criterion

In FP cavities, the resonance conditions of LG transverse modes are governed by the cavity geometry through the Gouy phase [19, 20]. In a stable two-mirror configuration, each LG mode accumulates a single-pass Gouy phase shift proportional to its transverse mode order $N = 2p + |l| + 1$, where p and l are the radial and azimuthal indices, respectively. Consequently, modes with distinct values of N exhibit distinct resonance frequencies within a given FSR. As the number of target modes grows, the cumulative Gouy phase shift for higher-order modes eventually exceeds one FSR, leading to a spectral wraparound effect in which the resonance frequencies of these modes revert toward the spectral positions originally occupied by lower-order modes.

To develop the design criterion in the simplest explicit form, we first consider a stable plane-

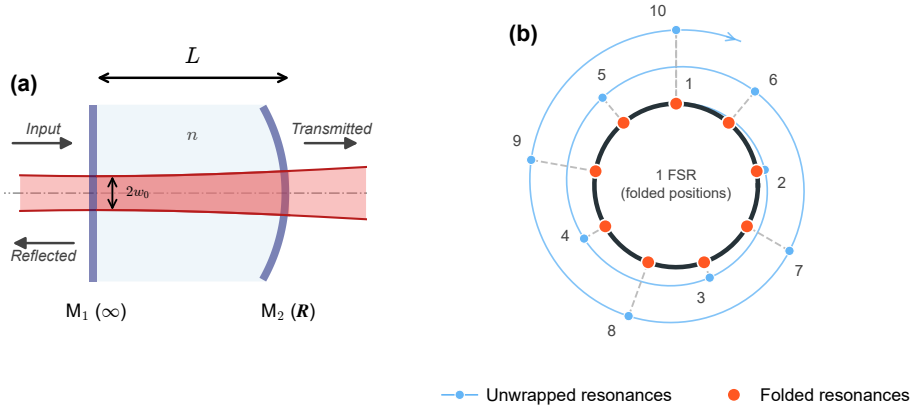


Fig. 1. Geometric origin of transverse-mode resonances and their single-FSR folded representation in a plane-concave cavity. (a) A plane-concave cavity (L , concave radius R) supports LG transverse modes with Gouy-phase-dependent resonance, enabling mode-selective spectral transmission while preserving the transverse field. (b) Archimedean-spiral representation of the single-FSR redistribution: mode N lies on the outer spiral at a radius proportional to the accumulated Gouy offset $(N-1)k$. Its radial projection onto the inner circle (spanning one FSR) yields the folded position $\text{pos}(N) = ((N-1)k) \bmod 1$ (orange dots). For the illustrative case $k = 2/9$ ($M = 9$), modes $N = 1-9$ ($l = 0-8$) land on nine uniformly spaced folded positions; mode $N = 10$ ($l = 9$) wraps back to the $N = 1$ neighborhood. The circular representation provides the geometric basis for defining the pairwise spacing s_{ij} and the minimum spacing $s_{\min}$ used in the unified design criterion.

concave FP cavity of length L and concave-mirror radius of curvature R , as shown in Fig. 1(a). In this configuration, the resonance mode distribution within one FSR is a fixed geometric property of the cavity, governed solely by the ratio L/R .

To systematically analyze this periodic structure, we reduce all resonance positions modulo one FSR and map them onto a unit circle, a procedure referred to as *spectral folding*. On this circle, each target mode corresponds to an angular position, and pairwise mode separations are represented as arc lengths, enabling direct comparison and optimization. Fig. 1(b) illustrates this mapping: the Gouy phase offset winds outward along an Archimedean spiral, and its radial projection onto the inner circle recovers the folded positions within one FSR. Based on this circular representation, we formulate a unified analytical model applicable to arbitrary finite sets of target LG transverse orders. The extension to more general stable two-mirror cavities, including symmetric double-concave cavities, is provided in Appendix C.

Quantitatively, for the plane-concave geometry under consideration, the resonance frequencies are given by

$$\nu_{q,N} = \text{FSR} \left(q + N \frac{\phi}{\pi} \right), \quad (1)$$

where q is the longitudinal mode index, $\phi = \arccos(\sqrt{1 - L/R})$ is the single-pass Gouy phase, $\text{FSR} = c/(2nL)$ is the free spectral range, and n is the intracavity refractive index. In terms of the normalized Gouy phase shift ϕ/π , which represents the transverse mode spacing relative to the FSR,

$$k \equiv \frac{\phi}{\pi} = \frac{1}{\pi} \arccos \left(\sqrt{1 - \frac{L}{R}} \right), \quad (2)$$

114 which depends only on the geometry ratio L/R , the normalized resonance position within one
 115 FSR consequently reduces to

$$\text{pos}(N) = (Nk) \bmod 1. \quad (3)$$

116 The longitudinal index q is eliminated by the modulo operation, so the folded arrangement
 117 depends solely on k . Shifting the origin to the fundamental mode yields the equivalent expression

$$\text{pos}(N) = ((N-1)k) \bmod 1. \quad (4)$$

118 This leaves all pairwise spacings invariant. Since k depends only on L/R , the folded arrangement
 119 remains unchanged under any tuning that preserves L/R , whether via wavelength, refractive
 120 index, or subwavelength cavity-length scanning.

121 By mapping the unit interval $[0, 1)$ onto a circle, each target mode is represented as a point.
 122 The design objective thereby reduces to maximizing the minimum angular separation among
 123 these points. For a target set of M modes (where M is the number of modes to be sorted), the
 124 relevant separation between modes N_i and N_j is given by the shorter arc length

$$s_{ij} = \min(|\text{pos}(N_i) - \text{pos}(N_j)|, 1 - |\text{pos}(N_i) - \text{pos}(N_j)|), \quad (5)$$

125 and the performance is limited by the minimum pairwise spacing

$$s_{\min} = \min_{i \neq j} s_{ij}. \quad (6)$$

126 The design task is to choose k (equivalently L/R) so as to maximize $s_{\min}$. An elementary bound
 127 constrains this maximum: M points partition the circle into M arcs whose lengths sum to unity,
 128 so

$$s_{\min} \leq \frac{1}{M}, \quad (7)$$

129 with equality if and only if the points are uniformly spaced. This geometric upper bound is
 130 universal, applying to every target set irrespective of its detailed structure. The degree to which a
 131 given set approaches this bound is determined by the algebraic properties of the order differences,
 132 as demonstrated by the two representative cases considered below.

133 2.1. Representative branch: consecutive and uniform-step target sets

134 For the consecutive target set $S_M = \{1, 2, \dots, M\}$, the folded positions given by $\{0, k, 2k, \dots,$
 135 $(M-1)k\} \bmod 1$ constitute an arithmetic progression on the circle. Consequently, uniform
 136 spacing of $1/M$ is attainable when k takes the fractional value m/M satisfying $\gcd(m, M) = 1$,
 137 yielding

$$k^* = \frac{m}{M}, \quad \gcd(m, M) = 1, \quad (L/R)^* = \sin^2\left(\frac{\pi m}{M}\right), \quad (8)$$

138 and saturating the geometric ceiling at $s_{\min} = 1/M$, as depicted in Fig. 2(a).

139 For $M = 9$, the coprimality condition yields three non-redundant branches, namely $m \in$
 140 $\{1, 2, 4\}$, considering the symmetry $m \leftrightarrow M-m$. Each branch attains the ideal spacing $1/9$. These
 141 branches, however, exhibit different sensitivities to fabrication tolerances. The geometry-to-step
 142 derivative can be expressed as

$$\frac{dk}{d(L/R)} = \frac{1}{2\pi\sqrt{(L/R)(1-L/R)}}.$$

143 It is minimized at $L/R = 0.5$. Consequently, branches for which the optimal geometry ratio
 144 $(L/R)^*$ is closer to the midpoint $L/R = 0.5$ display enhanced resilience to manufacturing
 145 uncertainties in the cavity geometry. Evaluation of this derivative at the three branch points

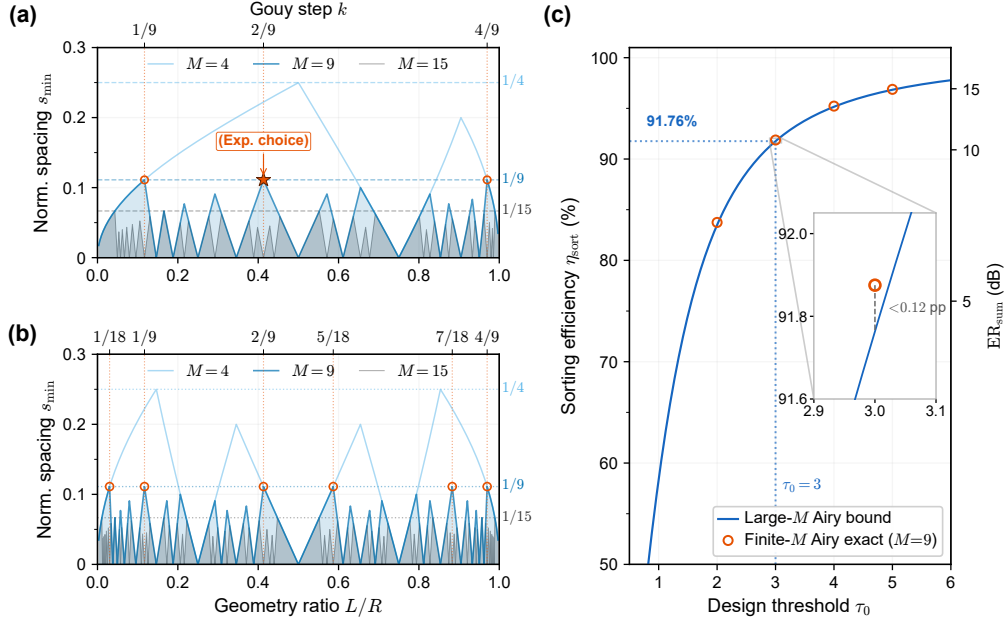


Fig. 2. Unified minimum-spacing criterion and its branch-specific design realizations. (a) Representative consecutive target sets: normalized minimum spacing $s_{\min}$ versus geometry ratio L/R for $M = 4, 9$, and 15 . For $M = 9$ the three coprime peaks ($m = 1, 2, 4$) each reach the ideal maximum of $1/9$; the star marks the adopted $m = 2$ branch ($k^* = 2/9$, $(L/R)^* \approx 0.4132$). (b) Uniform-step target sets with $\Delta_{\text{ord}} = 2$: same layout as (a) for $M = 4, 9$, and 15 . The coprime branches shift to $k = m/(2M)$ positions [Eq. (10)], but every curve still saturates its $1/M$ ceiling. For $M = 9$, the number of non-redundant branches increases from three to six, including the adopted $k^* = 2/9$ from the consecutive design. (c) Conservative sorting efficiency η_{sort} and system extinction ratio ER_{sum} versus design threshold τ_0 in the large- M Airy limit [Eq. (16)]. Orange open circles are the exact finite- M Airy predictions for $M = 9$ at $\tau_0 = 2$ – 5 , all within 0.5 percentage points of the bound (see inset). The dotted crosshair marks the adopted $\tau_0 = 3$, corresponding to $\eta_{\text{sort}} = 91.76\%$ and $\text{ER}_{\text{sum}} > 10.47$ dB.

146 yields $dk/d(L/R) = 0.495$ for $m = 1$, corresponding to $(L/R)^* \approx 0.117$; 0.323 for $m = 2$,
 147 corresponding to $(L/R)^* \approx 0.413$; and 0.931 for $m = 4$, corresponding to $(L/R)^* \approx 0.970$.
 148 Thus, for a fixed tolerance in L/R , the resulting shifts in k are $1.5\times$ and $2.9\times$ larger for the $m = 1$
 149 and $m = 4$ branches, respectively, relative to the observed minimum at $m = 2$. The $m = 2$ branch
 150 is adopted for the experiment reported below, for its combination of optimal spacing and the
 151 lowest sensitivity among the three branches.

152 The consecutive case represents the simplest instance of a more general uniform-step family.
 153 When the target mode indices follow an arithmetic progression $N_i = N_0 + (i - 1)\Delta_{\text{ord}}$, a change
 154 in the starting index N_0 merely results in a global rotation of the points on the circle. The step
 155 size Δ_{ord} , in contrast, defines an effective circular increment,

$$u = (\Delta_{\text{ord}} k) \bmod 1. \quad (9)$$

156 In terms of u , a uniform-step set of M modes is equivalent to an M -mode consecutive problem.
 157 The geometric ceiling $s_{\min} = 1/M$ is therefore attained when the following coprimality condition
 158 is satisfied:

$$\Delta_{\text{ord}} k \equiv \frac{m}{M} \pmod{1}, \quad \gcd(m, M) = 1. \quad (10)$$

159 The corresponding cavity geometry is given by $L/R = \sin^2(\pi k)$. The entire uniform-step family
 160 thus saturates the $1/M$ bound. This is illustrated in Fig. 2(b) for $\Delta_{\text{ord}} = 2$: for $M = 9$, the
 161 landscape now admits six coprime branches reaching $s_{\min} = 1/9$, compared with only three in
 162 the consecutive case, while $M = 4$ and $M = 15$ retain their respective ceilings. The adopted
 163 geometry $k^* = 2/9$ remains optimal for both step sizes.

164 2.2. Extension to general target sets

165 The model is not restricted to uniform-step progressions. For a general target set $S = \{N_1, \dots,$
 166 $N_M\}$ with pairwise order differences $D(S) = \{N_j - N_i : i < j\}$, the minimum spacing takes the
 167 form

$$s_{\min}(k; S) = \min_{d \in D(S)} \|dk\|, \quad (11)$$

168 where $\|x\|$ denotes the distance from x to its nearest integer. Over any rational search window
 169 $I = [a, b]$, this function is the piecewise-linear lower envelope of finitely many V-shaped profiles
 170 $\|dk - n\|$, so its global maximum is attained at a rational value $k^* = m/q$ whose denominator
 171 satisfies an explicit bound

$$q \leq Q_I(S) = \max(\text{den}(a), \text{den}(b), 2D_{\max}), \quad D_{\max} = \max D(S). \quad (12)$$

172 where $\text{den}(\cdot)$ denotes the denominator of a rational number written in lowest terms. The design
 173 problem therefore reduces to an exact search over finitely many rational candidates (m, q) with
 174 $\gcd(m, q) = 1$ and $a \leq m/q \leq b$. For uniform-step sets, this recovers the closed-form result of
 175 the preceding subsection; for general finite sets, it provides the corresponding optimum without a
 176 continuous scan. A proof is provided in Appendix A.2.

177 2.3. Resolvability criterion and unified capacity boundary

178 So far the analysis has established how to maximize the minimum spacing $s_{\min}$ for a given target
 179 set. To translate this geometric spacing into a physical crosstalk level, it must be compared with
 180 the spectral width of the cavity transmission peak.

181 Each resonance of the FP cavity has a full width at half maximum $\text{FWHM} = \text{FSR}/\mathcal{F}$, where
 182 $\mathcal{F}$ is the cavity finesse. The physical frequency gap between two modes with circular spacing s_{ij}
 183 is $\Delta\nu_{ij} = s_{ij} \cdot \text{FSR}$. Dividing this frequency gap by the linewidth yields their separation in units
 184 of linewidths:

$$\tau_{ij} \equiv \frac{\Delta\nu_{ij}}{\text{FWHM}} = \mathcal{F} \cdot s_{ij}, \quad \tau_{\min} = \mathcal{F} \cdot s_{\min}. \quad (13)$$

185 τ counts how many linewidths separate two peaks. The pair with the smallest τ exhibits the
 186 highest crosstalk, and this bottleneck value is denoted $\tau_{\min}$. To impose a design requirement
 187 on mode isolation, we introduce a threshold τ_0 , defined as the minimum acceptable separation
 188 between any two resonances in units of linewidths. The crosstalk at a given τ is governed by the
 189 cavity transmission lineshape. The ideal FP transmittance follows the Airy function,

$$T(\nu) = \frac{1}{1 + \left(\frac{2\mathcal{F}}{\pi}\right)^2 \sin^2\left(\frac{\pi\nu}{\text{FSR}}\right)}. \quad (14)$$

190 When the cavity is locked to a target mode, any non-target mode detuned by $\Delta\nu$ still transmits a
 191 fraction $T(\Delta\nu)$. This residual transmission constitutes the physical origin of inter-mode crosstalk.

192 For M uniformly spaced target modes operating at the capacity-boundary finesse $\mathcal{F} = M\tau_0$,
 193 the total crosstalk associated with any target mode is the exact Airy circular sum

$$P_{\text{noise}}^{(M)}(\tau_0) = \sum_{n=1}^{M-1} \frac{1}{1 + \left(\frac{2M\tau_0}{\pi}\right)^2 \sin^2\left(\frac{\pi n}{M}\right)}. \quad (15)$$

194 This sum admits a closed-form expression (see Appendix A.1) and is a monotonically increasing
 195 function of M . In the asymptotic limit $M \rightarrow \infty$ with τ_0 held fixed, it converges to

$$P_{\text{noise}}^{(\infty)}(\tau_0) = \frac{\pi}{2\tau_0} \coth \frac{\pi}{2\tau_0} - 1, \quad (16)$$

196 which is independent of M and serves as a conservative upper bound on the crosstalk: for any
 197 finite M , $P_{\text{noise}}^{(M)} < P_{\text{noise}}^{(\infty)}$. The corresponding sorting efficiency and extinction ratio are given by

$$\text{ER}_{\text{sum}}^{(\infty)} = 10 \lg \frac{1}{P_{\text{noise}}^{(\infty)}}, \quad \eta_{\text{sort}}^{(\infty)} = \frac{1}{1 + P_{\text{noise}}^{(\infty)}}. \quad (17)$$

198 Taking $M = 9$ at $\mathcal{F} = 27$ as an example, the exact finite- M prediction gives an extinction ratio of
 199 $\text{ER}_{\text{sum}}^{(9)} = 10.53$ dB, a mere 0.06 dB above the conservative bound [see Fig. 2(c)].

200 The design threshold τ_0 specifies the minimum acceptable spacing between any two cavity
 201 resonances in units of linewidths. By imposing $\tau_{\text{min}} \geq \tau_0$, the geometric frequency separation
 202 directly translates into a guaranteed isolation level. The relationship between τ_0 and the
 203 corresponding conservative performance bounds is shown in Fig. 2(c); numerical values for
 204 several representative thresholds are summarized in Table 1. For the threshold $\tau_0 = 3$ adopted
 205 throughout this work, we obtain $\eta_{\text{sort}}^{(\infty)} = 91.76\%$ and $\text{ER}_{\text{sum}}^{(\infty)} > 10.47$ dB.

Table 1. Conservative lower bounds on sorting performance.

τ_0	$\text{ER}_{\text{sum}}^{(\infty)}$ (dB)	$\eta_{\text{sort}}^{(\infty)}$ (%)
1.0	1.47	58.39
2.0	7.04	83.50
3.0	10.47	91.76
4.0	12.93	95.16
5.0	14.86	96.84
6.0	16.43	97.78

Values are the asymptotic extinction ratio $\text{ER}_{\text{sum}}^{(\infty)}$ and sorting efficiency $\eta_{\text{sort}}^{(\infty)}$ for representative design thresholds τ_0 , evaluated from the large- M limit of the exact Airy circular sum. Closed-form expressions are given in Eqs. (16–17).

206 Combining the threshold condition $\tau_{\text{min}} \geq \tau_0$ with the universal geometric upper bound
 207 $s_{\text{min}} \leq 1/M$ leads directly to the central design inequality:

$$M \leq \left\lfloor \frac{\mathcal{F}}{\tau_0} \right\rfloor. \quad (18)$$

208 This bound applies universally to any target set and any cavity geometry: a single FP cavity
 209 of finesse $\mathcal{F}$ can sort at most $\lfloor \mathcal{F}/\tau_0 \rfloor$ modes at the required isolation level. For uniform-step
 210 target sets, the optimal arrangement attains the bound with $s_{\text{min}} = 1/M$. For nonuniform target
 211 sets, equality in this bound is not guaranteed: it depends on whether the set can be mapped by
 212 an admissible cavity step to an equally spaced circular arrangement. When this condition is
 213 not met, the optimum has $s_{\text{min}}^* < 1/M$, and a higher finesse is required to compensate for the
 214 resulting spacing deficit. While the analysis uses a plane-concave cavity, the model depends
 215 on k . For a general stable two-mirror cavity, the step generalizes to $k_{\text{eff}} = \pi^{-1} \arccos(\sqrt{g_1 g_2})$
 216 with $g_{1,2} = 1 - L/R_{1,2}$; all results carry over (see Appendix C).

217 3. Nine-Mode Boundary-Probing Validation

218 This section experimentally validates the proposed model on a representative consecutive-mode
 219 boundary. The target set consists of $M = 9$ consecutive LG modes with $p = 0$ ($l = 0-8$). For this
 220 set, the closed-form optimum [Eq. (8)] yields $k^* = 2/9$ and $(L/R)^* \approx 0.4132$. In this subspace,
 221 $N = |l| + 1$ gives a one-to-one correspondence between the mode label and the transverse order.
 222 The folded ordering and the first return beyond the target set can therefore be read directly from
 223 measured spectra. At this operating point, the nine target resonances are equally spaced around
 224 the single-FSR circle with a uniform spacing of $1/9$ of one FSR, whereas the next-order mode
 225 with $l = 9$ (i.e., $N = 10$) returns to the neighborhood of the fundamental. This return signifies
 226 the predicted onset of spectral overlap at the nine-mode capacity boundary. For the design
 227 threshold $\tau_0 = 3$, the capacity inequality [Eq. (18)] imposes a minimum finesse requirement of
 228 $\mathcal{F}_{\min} = M\tau_0 = 27$.

229 The measurement campaign establishes three key aspects of the model: (i) the folded resonance
 230 arrangement from the single-FSR circular representation; (ii) the crosstalk structure governed
 231 by the minimum folded spacing $s_{\min}$; (iii) the quantitative performance bounds set by the τ
 232 criterion. In the subsequent subsections, we first demonstrate that the fabricated cavity operates
 233 at the intended design point, followed by a verification of the folded spectral arrangement and
 234 the next-mode return. We conclude with an assessment of the sorting performance under an
 235 equal-weight superposition of the nine modes as input.

236 3.1. Experimental setup

237 The optical layout is shown in Fig. 3. A narrow-linewidth tunable laser centered at 795 nm
 238 passes through a $(2.5\times)$ beam expander. The expanded beam is then addressed by a reflective
 239 phase-only spatial light modulator (SLM1), which imprints the required complex amplitude
 240 via holography [21] to produce the nine target LG_0^l modes ($l = 0-8$). Two successive $4f$
 241 relays perform three functions: selecting the first-order diffraction, providing spatial filtering,
 242 and matching the beam waist to the cavity mode. The prepared beam is then coupled into a
 243 temperature-stabilized fused-silica plane-concave FP cavity. The geometry ratio of this cavity is
 244 set near $(L/R)^* \approx 0.4132$.

245 Two measurement configurations share the same input path. For spectral characterization,
 246 the laser frequency is swept over ≈ 14.2 GHz, and the cavity transmission is recorded by a
 247 photodetector. For the equal-weight nine-mode superposition measurement, the second spatial
 248 light modulator (SLM2) imprints conjugate holograms onto the transmitted field. The shaped
 249 field is then coupled into a single-mode fiber via a $4f$ relay lens system.

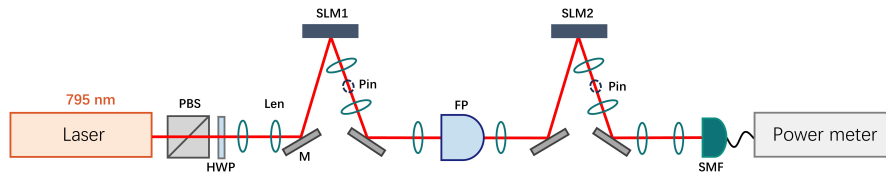


Fig. 3. Experimental setup for the nine-mode FP-cavity sorter. A tunable laser (795 nm) is beam-expanded ($2.5\times$) and encoded onto a reflective phase-only SLM1 via complex-amplitude holography to generate the target LG_0^l modes ($l = 0-8$). Relay optics provide first-order selection, spatial filtering, and cavity-mode matching into a temperature-stabilized plane-concave FP cavity. The schematic shows the common input path and the mode-resolved projection arm. For mode-resolved measurements with an equal-weight superposition input, the transmitted field is projected by SLM2, relayed to a single-mode fiber, and recorded by a power meter. For spectral characterization, the cavity transmission is recorded by a photodetector at the transmission port.

250 3.2. Operating-point calibration

251 Frequency-domain transmission measurements confirm that the fabricated cavity operates close
 252 to the predicted design point. The measured FSR and finesse are 9.915 ± 0.004 GHz and
 253 $\mathcal{F} = 32.21 \pm 0.19$, respectively. A fit to the resonance positions yields a normalized Gouy
 254 step of $k = 0.2228 \pm 0.0003$, within 0.3% of the design value $k^* = 2/9 \approx 0.2222$. Using the
 255 plane-concave relation between k and L/R , this fitted step corresponds to an effective geometry
 256 ratio $(L/R)_{\text{eff}} \approx 0.4146$, close to the target value $(L/R)^* \approx 0.4132$.

257 Using the nine measured target resonances, the minimum folded pairwise spacing is $s_{\text{min}}^{\text{meas}} \approx$
 258 0.1086 . The corresponding linewidth-scaled resolvability is $\tau_{\text{min}}^{\text{meas}} = \mathcal{F} s_{\text{min}}^{\text{meas}} \approx 3.50$ [Eq. (13)],
 259 placing the device above the design threshold $\tau_0 = 3$. At the threshold finesse $\mathcal{F}_{\text{min}} = 27$,
 260 satisfying this condition would require the nine-mode spacing to attain the geometric upper bound
 261 $s_{\text{min}} = 1/9$, leaving no spacing margin against deviations of k from an exact optimal branch.
 262 The measured finesse exceeds this threshold by $\sim 19\%$. This excess finesse provides a tolerance
 263 window of $|\Delta k| \lesssim 4 \times 10^{-3}$ around the selected design point, as quantified in Appendix B.2.

264 3.3. Spectral verification and next-mode return

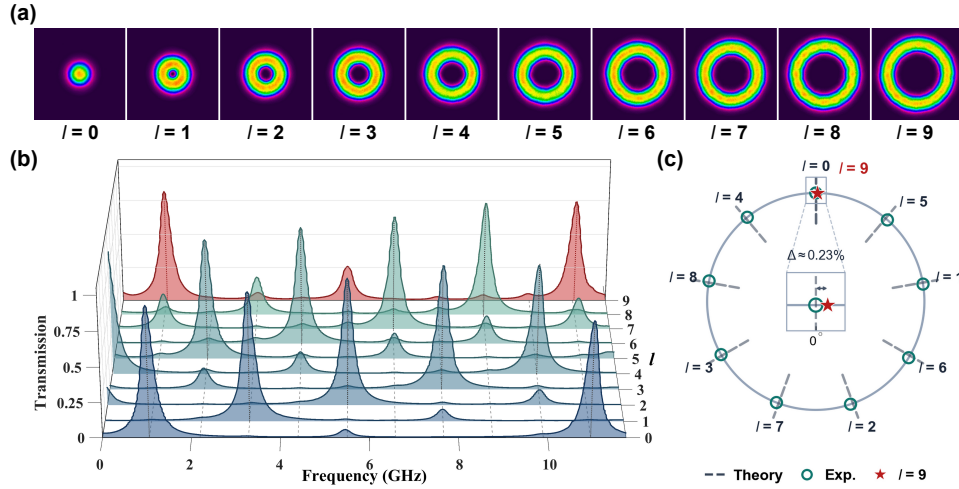


Fig. 4. Experimental validation of the representative consecutive-mode boundary case. (a) Charge-coupled device (CCD) images of cavity-transmitted individual OAM modes at their respective resonance peaks ($l = 0$ –9), confirming transverse-profile preservation through the FP cavity. (b) Measured transmission spectra of $l = 0$ –9 within one FSR, showing the non-monotonic peak arrangement predicted by the folding model at the consecutive optimum. The $l = 9$ resonance (red) returns to the $l = 0$ neighborhood at the FSR boundary. Weak satellite resonances in several higher- l traces are consistent with residual $p = 1$ radial-mode excitation. (c) Polar representation of measured peak positions (circles) versus theoretical predictions (dashed, $k^* = 2/9$). Inset: magnified next-mode return region showing a residual separation of 0.23% FSR.

265 The cavity-transmitted images recorded at the resonance peaks [see Fig. 4(a)] retain the
 266 expected OAM-dependent transverse structures of the target modes, confirming that the FP
 267 cavity preserves the transverse mode profile during sorting. The measured individual-mode
 268 transmission spectra [as shown in Fig. 4(b)] provide a direct test of the folding prediction: within
 269 one FSR, the nine target resonances do not follow the monotonic order of the mode index but
 270 instead appear at the positions prescribed by $\text{pos}(N) = ((N-1)k) \bmod 1$ [Eq. (4)]. The average
 271 frequency deviation between measured and predicted peak positions is ≈ 28 MHz, less than

one-tenth of the measured linewidth (≈ 307 MHz). The polar comparison in Fig. 4(c) confirms this quantitative agreement across all nine target modes.

Weak satellite resonances visible in several higher- l traces are consistent with residual $p = 1$ radial-mode excitation from imperfect cavity-mode matching. For fixed l , this component is shifted by two transverse-order units relative to the intended $p = 0$ mode, as follows from the mode index relation $N = 2p + |l| + 1$.

The folding model also predicts the spectral position of the first mode outside the target set. For $l = 9$ ($N = 10$), the folding formula gives $\text{pos}(10) = ((10-1) \times 2/9) \bmod 1 = 0$, placing this mode at the same folded position as the fundamental. The measured $l = 9$ resonance [red curve in Fig. 4(b); red star in Fig. 4(c)] falls only 23 MHz (≈ 0.07 linewidths, 0.23% FSR) from the $l = 0$ resonance, confirming this predicted return at the nine-mode capacity boundary.

3.4. Sorting performance under simultaneous nine-mode input

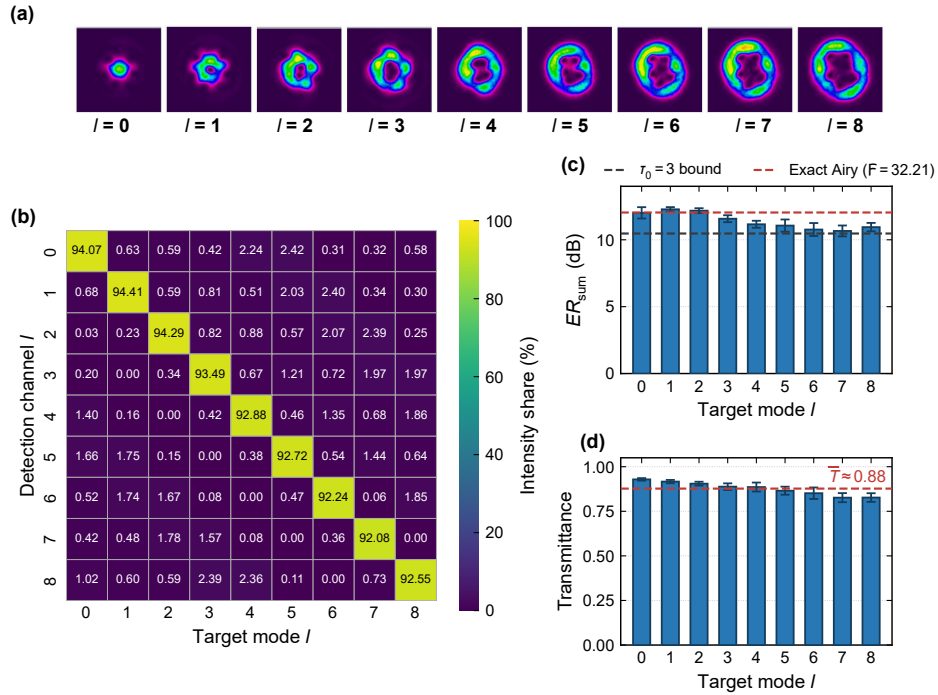


Fig. 5. Sorting under simultaneous nine-mode input. (a) CCD spot patterns recorded at the $l = 0$ – 8 resonance peaks. (b) Measured 9×9 normalized mode-sorting matrix with an equal-weight superposition input; diagonal entries indicate sorting efficiencies exceeding 92%. (c) Per-mode system extinction ratio (ER_{sum}) compared with the conservative large- M lower bound (10.47 dB) and the exact finite- M Airy prediction (12.04 dB). (d) Single-mode peak transmittance for $l = 0$ – 8 . The gradual decrease with increasing l contributes to the lower extinction ratios of the higher- l modes in (c).

To evaluate the FP-cavity sorting performance in the nine-mode LG_0^l target basis, we injected an equal-weight superposition of all nine target modes into the cavity and read out the transmitted field by sequential projection using the projection arm consisting of SLM2 and a single-mode fiber. This projection arm readout uses conjugate complex-amplitude holograms [21] followed by single-mode-fiber coupling, which can introduce mode-dependent readout efficiencies and finite projection crosstalk. We independently characterized the projection arm using single-mode inputs.

We obtained a readout-response matrix and used this matrix to account for the readout response in the raw projection vectors before column-wise normalization [22, 23]. The resulting normalized mode-sorting matrix [see Fig. 5(b)] is strongly diagonally dominant, with sorting efficiencies ranging from 92.08% to 94.41% and a mean sorting efficiency of $93.19\% \pm 0.37\%$ (95% confidence interval from three independent runs). Residual off-diagonal leakage is concentrated among nearest-neighbor mode pairs on the folded circle, consistent with the minimum-spacing bottleneck predicted by the model. Further details of the response-matrix procedure are provided in Appendix B.1.

The mode-resolved extinction ratios [Fig. 5(c)] provide the most direct quantitative comparison with the model predictions. The model establishes two reference levels: a conservative lower bound of 10.47 dB set by the design threshold $\tau_0 = 3$ in the large- M limit (Table 1), and an exact finite- M Airy prediction of 12.04 dB evaluated at the measured finesse under equal spacing. The measured mean extinction ratio is 11.41 ± 0.25 dB, and the worst-performing mode ($l = 7$) is 10.66 dB. All modes fall above the conservative lower bound and below the ideal Airy reference level, placing the measured performance within the design regime predicted by the model. The single-mode peak transmittance in Fig. 5(d) decreases gradually with increasing azimuthal index, with a mean value of $\bar{T} \approx 0.88$. This trend is consistent with the larger spatial extent of higher- l modes, which increases their sensitivity to finite aperture, cavity-mode matching, and input-alignment errors. It also helps explain the per-mode variation in Fig. 5(c): for higher- l target modes, the transmitted signal is weaker, whereas leakage from folded lower- l neighbors on the folded circle can have higher peak transmittance.

Together, the operating-point calibration, the folded resonance arrangement, the next-mode return at $l = 9$, and the nine-mode sorting performance complete the three-part experimental validation outlined at the beginning of this section. Design extensions and the broader applicability of the model are discussed in the following section.

4. Discussion

The preceding experiment validates the model at the nine-mode capacity boundary. The same capacity equation sets the design scale for other mode counts: $\mathcal{F}_{\min} = M\tau_0$ increases linearly with M , so each additional target mode requires only τ_0 units of additional finesse. Table 2 lists the resulting design blueprint from $M = 4$ to $M = 100$ at $\tau_0 = 3$, including the preferred geometry branch, the corresponding L/R , and the predicted performance for each case. The predicted efficiency converges rapidly to the large- M limit, whereas the extinction ratio varies only weakly across the table.

Table 2. Consecutive-set sorting design blueprint for $\tau_0 = 3$.

M	$\mathcal{F}_{\min}$	m	k^*	$(L/R)^*$	$\text{ER}_{\text{sum}}^{(M)}$ (dB)	$\eta_{\text{sort}}^{(M)}$ (%)
4	12	1	1/4	0.5000	10.80	92.3
9	27	2	2/9	0.4132	10.53	91.9
15	45	4	4/15	0.5523	10.49	91.8
30	90	7	7/30	0.4477	10.48	91.8
50	150	13	13/50	0.5314	10.47	91.8
100	300	23	23/100	0.4373	10.47	91.8

Here $\mathcal{F}_{\min} = 3M$; the preferred m is the coprime branch nearest the lowest-sensitivity geometry. ER and η are finite- M Airy predictions at $\mathcal{F} = \mathcal{F}_{\min}$; formulas are given in Appendix A.1.

The present validation uses the $p=0$ subspace, where the transverse order $N = |l| + 1$ gives a one-to-one correspondence between the mode label and N and makes the folding sequence directly observable. The design model is written for the general transverse order $N = 2p + |l| + 1$, and FP-based mode selectivity involving multiple radial orders has been demonstrated in related configurations [16]. The same spacing criterion also determines the design of other finite target sets: for uniform-step target sets, equality in the capacity bound is attainable [Eq. (10)], whereas for general nonuniform sets, the best geometry follows from the exact rational search of Section 2. Since the analysis depends only on the normalized Gouy step k , it is not tied to the plane-concave cavity used in the experiment. The folding analysis, crosstalk sums, and capacity bound carry over to general stable two-mirror cavities via the effective Gouy step defined in Appendix C. Different cavity types, such as symmetric double-concave cavities, modify only the geometry-to-step mapping and the branch-sensitivity ranking; the capacity condition itself remains unchanged.

At the cavity-design level, the relevant parameters are the normalized Gouy step k and the finesse $\mathcal{F}$. Because the sorting mechanism is a passive linear spectral response of the cavity, the same transfer function applies to single-photon inputs, provided the photon bandwidth remains narrow relative to the cavity linewidth. Related work has demonstrated Gouy-phase-controlled evolution in multi-photon quantum states [24]. The capacity boundary and design rules derived here therefore provide quantitative guidance for extending FP-based mode sorting toward high-dimensional quantum information processing [4, 6].

This dimensionless structure also clarifies how the design scales. Within the model, preserving the effective Gouy step and the finesse preserves the normalized sorting performance, independent of the absolute cavity size. This property enables compact and integrated implementations. Integrated lithium-niobate FP micro-resonators, for instance, provide a compact FP platform with electro-optic intracavity refractive-index control for resonance tuning [25]. The present model specifies the effective Gouy step and finesse required for a target sorting capacity, thereby providing design guidance for such miniaturized FP platforms.

5. Conclusion

This work establishes an analytical model based on spectral folding for high-dimensional LG mode sorting in FP cavities. The single-FSR circular representation reduces cavity design to a one-parameter optimization on the normalized Gouy step k . Together with the finesse, the minimum folded spacing defines the resolvability criterion $\tau_{\min} = \mathcal{F} s_{\min} \geq \tau_0$, which translates geometric spacing into a guaranteed isolation level and yields the capacity boundary $M \leq \lfloor \mathcal{F}/\tau_0 \rfloor$ for arbitrary finite target sets. A nine-mode boundary experiment quantitatively validates the model: measured resonance positions agree with predictions to within 0.1 linewidth, the next mode returns to the fundamental's spectral neighborhood at the predicted capacity boundary, and sorting under an equal-weight superposition of all nine modes yields a mean efficiency of 93.19% and a mean extinction ratio of 11.41 dB, with all target modes falling within the predicted design regime. Because the cavity response is governed by the normalized Gouy step and the finesse, the model extends to stable two-mirror cavities through the effective Gouy step and provides a quantitative analytical basis for the design and performance prediction of high-dimensional FP mode sorters.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

371 **Supplemental document.** See Supplement 1 for supporting content.

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